Final Design Project

ENGN 0720: Thermodynamics

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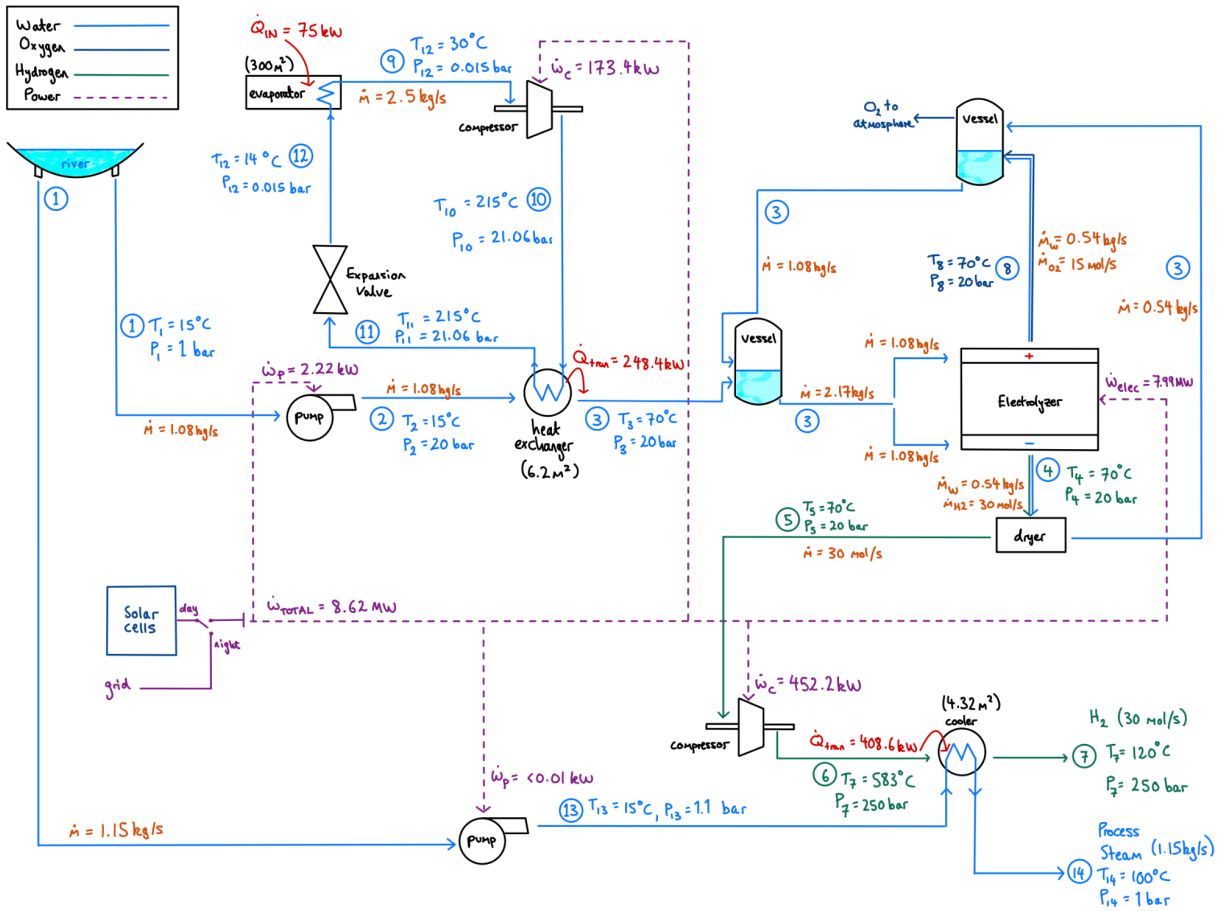

Design Objective

Provide 30 mol s^{-1} dry H_2 at 250 bar and < 120 °C to a nearby ammonia plant, operating continuously on 12 h daylight solar power and 12 h electrical grid power at night. Low-pressure steam can be marketed if it is economical.

Process Design

The final configuration consists of four functional blocks:

1. High-pressure electrolysis (20 bar, 70 °C). An alkaline stack rated for 7.79 MW produces H_2/O_2 from liquid water at 70 °C. Internal air cooling (provided by the vendor) removes overpotential heat.

2. Gas clean-up and drying. The cathode outlet gas–liquid separator delivers saturated H_2 via a sorbent dryer (only operating cost) to meet the "dry" requirement.

3. Compression & after-cooling. H_2 is increased from 20 → 250 bar by a single-stage compressor (η = 70 %); a shell-and-tube exchanger cools the discharge to 120 °C across a 4.32 m² area.

4. Utility/heat-recovery loop. River water at 15 °C is pressurized to 20 bar, raised to 70 °C to feed the stack, and finally passed through the compressor's after-cooler to recover 0.41 MW as 1-bar, 100 °C steam to be sold to the ammonia factory.

Energy Strategy

Rather than CAPEX-expensive pumped hydro, we install 8.62 MW of Si-PV (\$3.20 W^{-1}) to provide the entire day-time electrical load (electrolyser + compressor + pumps ≈ 8.25 MW). The grid supply meets night load (12 h) at \$0.20 kWh^{-1}. This hybrid option reduces both CAPEX and operating costs.

Thermodynamic & operability checks

All rotating equipment within vendor limits (compressor T max= 283 °C < 300 °C). First- and second-law balances affirm no violations; overall exergy destruction is dominated by the electro-chemical overpotential, unavoidable for current technology.

Design Philosophy

Each choice was measured against one criterion at every step: does this improvement result in a simpler, less energy-intensive plant without sacrificing performance specifications? This evaluation procedure led to a 20-bar and 70 °C high-pressure alkaline electrolyzer driven by a single compressor, instead of two or three rotary machines, thus guaranteeing a compact machine arrangement that is less maintainable and has potential for automation. Then, we carefully considered compliance with the simplicity criterion for each ancillary concept that arose. For instance, while a Joule–Thomson expansion valve was at first thought to be advantageous in cooling the 250-bar hydrogen stream, both first-law and exergy analyses, as well as Professor Hunt's comments during the design review, revealed that the throttling loss was significant compared to the small refrigeration gain obtained. Elimination of this valve not only avoided a potential reliability issue but also allowed diversion of the compressor discharge at 283 °C to a recuperative heat exchanger, which now heats 0.41 MW of process water to 100 °C while creating a 1-bar steam by-product worth about \$0.5 million per year.

The same philosophy of "optimizing waste" dictated our inclusion of solar panel heat recovery. Since crystalline-silicon solar cells convert only about 20% of light input to electricity,

the remaining 80% is released as low-grade heat, commonly vented to the atmosphere. By passing water across the backs of the panels, which are at 80 °C, we recover this excess energy, which is then, in turn, used to preheat the feed to the electrolyzer, substituting for electrical heating without the added equipment—the system uses the same pumps and heat exchangers that the compressor after-cooler utilizes. Every modification was rooted in a common design philosophy: to maximize passive, inherent synergies over add-ons, minimize exergy losses at their origin, and exploit any heat or pressure dissipation that cannot be avoided. The final design conforms to this philosophy—having a single compressor, no throttling valves, and the complete recycle of every watt of exhaust heat—yielding a vigorous, user-friendly plant that meets 30 mol/s H_2 demand at an effective internal rate of return of about 9.7%.

Economic Analysis

Capital Investment (Note: for sake of analysis, all CapEx is assumed at t = 0)

The depreciation of this analysis is assuming that we are keeping with the GAAP regulations mandated method of straight line depreciation of the useful lives of the plants CAPEX over the 10 years we have assumed operation for.

Item	***Basis***	***Cost per Unit***	***Cost (M $)***
PV Array	8.62 MW_P	3.20 $/W	27.58
Electrolyster Stacks	7.79 MW	40 $/kW	0.31
H_2 Compressor	452 kW_E	0.5 $/W	0.23
Pumps, Piping, Controls & HX	–	–	0.55

Total CapEx ····· 31.3 M $

Annual Operating Assumptions:

In reality, this plant will not be active for 24 hours a day, 365 days a year, to take this into consideration we assumed an annual downtime of 5%.

Item	***Value***
Operating Hours	0.95 x 8760 = 8322 hours
H_2 Production	30 mol/s x 8322 hours = 898,776 kmol/yr
1-Bar Steam Co-Products	1.15 kg/s x 8322 hours = 34.50 kt/yr
Grid-power hours (Night only)	12 h/d x 365 d/yr x 0.95 = 4161 hours

Night Electrical Load (including electrolyser)	8.25 MW

Annual Revenues:

Item	***Calculation***	***Cost (M $ / Yr)***
Grid Electricity	8.25 MW x 4161 h x 0.2 $ kW/H	6.87
Sorbent Replacement	(0.54 kg/s H_2O x 8332 h x 3600 s/h)/(18 kg kmol/$ kmol)	0.90
General O&M	5% of CapEx	1.57
Total		9.34

Income Statement and Free Cash Flow for the First Year:

EBITDA	14.90 - 9.34 = 5.56
Depreciation	3.13
EBIT	2.43
Tax (20% corporate tax rate assumed)	0.49
Net Income	**1.95**
Depreciation Add-Back (Non-Cash expense)	+ 3.13
Free Cash Flow	**5.08 M $/yr**

NPV and IRR Calculations:

For the purposes of this project we assumed a 4% real discount rate on our taxes each year for the NPV calculator, as well as that the funding for this plant would be unleavened meaning the operating/owning company took on no debt to create this plant. The final assumption we operated with for this aspect of the calculations was the no salvage value at the end of the 10 years meaning the plant won't be worth anything at the end of its life.

$$PV\ Factor = \frac{1-(1+0.04)^{-10}}{0.04} = 8.11$$

$$NPV = -31.3 + (8.05 \times 8.11) = +9.9\ M\$$$

These numbers correspond with an internal rate of return of around 9.7%, which is around the typical IRR for real estate and plant investment.

Final Cash Flow Summary:

Metric	*Year 0*	*Year 10*
Free Cash Flow (M $)	-31.3	+5.08
Discount Factor at 4%	1	0.676
Present Value of FCF	-31.3	+41.3
Cumulative PV	**-31.3**	**+9.9 (at end of life)**

Conclusion

Our final configuration meets both technical and financial targets. Operating the electrolyser at 20 bar trims compression work by roughly 30 %, so a single-stage compressor stays below the 300 °C discharge limit, while the recuperative after-cooler captures 0.41 MW of otherwise wasted heat and converts it into 1-bar steam (≈ 1.15 kg s^{-1}) that earns about $0.52 million per year. In an early iteration we inserted a Joule–Thomson expansion valve to cool the 250 bar hydrogen, but throttling squandered exergy and provided only marginal refrigeration; replacing the valve with the heat-recovery loop removed a failure-prone component and boosted annual profit by roughly $0.5 million. Coupled with a photovoltaic array sized to cover the full daylight load and inexpensive grid electricity at night—a combination that still outperforms pumped-hydro storage on a life-cycle basis—the plant now delivers an NPV of about $10 million and a real IRR of ~9.7 %, all while supplying zero-carbon hydrogen to the neighbouring ammonia facility